

Activity 4

Card Flip Magic—*Error Detection & Correction*

Summary

When data is stored on a disk or transmitted from one computer to another, we usually assume that it doesn't get changed in the process. But sometimes things go wrong and the data is changed accidentally. This activity uses a magic trick to show how to detect when data has been corrupted, and to correct it.

Curriculum Links

- ✓ Mathematics: Number – Exploring computation and estimation.
- ✓ Mathematics: Algebra – Exploring patterns and relationships, solving for a missing value.
- ✓ Mathematics: Rows and columns, coordinates
- ✓ Technology: Validating data

Skills

- ✓ Counting
- ✓ Recognition of odd and even numbers

Ages

- ✓ 7 years and up

Materials

- ✓ A set of 36 “fridge magnet” cards, coloured on one side only
- ✓ A metal board (a whiteboard works well) for the demonstration.

Each pair of students will need:

- ✓ 36 identical cards, coloured on one side only.

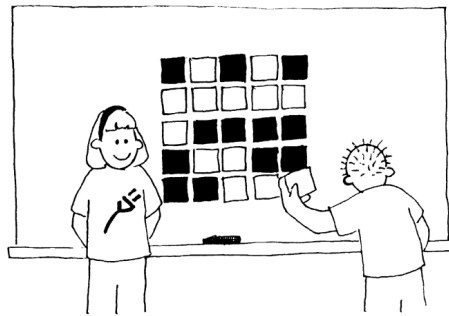
The “Magic Trick”

Demonstration

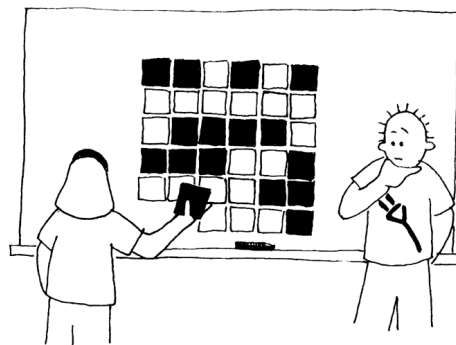
Here’s your chance to be a magician!

You will need a pile of identical, two-sided cards. (To make your own cut up a large sheet of card that is coloured on one side only). For the demonstration it is easiest to use flat magnetic cards that have a different colour on each side—fridge magnets are ideal, but make sure they are magnetic on both sides (many are single sided, in which case you can glue them face to face, and put a white dot on one side).

1. Choose a student to lay out the cards in a 5×5 square, with a random mixture of sides showing.



Casually add another row and column, “just to make it a bit harder”.



These cards are the key to the trick. You must choose the extra cards to ensure that there is an even number of coloured cards in each row and column.

2. Get a student to flip over one card only while you cover your eyes. The row and column containing the changed card will now have an odd number of coloured cards, and this will identify the changed card. Can the students guess how the trick is done?

Teach the trick to the students:

1. Working in pairs, the students lay out their cards 5×5 .
2. How many coloured cards are there in each row and column? Is it an odd or even number? Remember, 0 is an even number.
 3. Now add a sixth card to each row, making sure the number of coloured cards is always even. This extra card is called a “parity” card.
 4. Add a sixth row of cards along the bottom, to make the number of cards in each column an even number.
 5. Now flip a card. What do you notice about the row and column? (They will have an odd number of coloured cards.) Parity cards are used to show you when a mistake has been made.
 6. Now take turns to perform the ‘trick’.

Extension Activities:

1. Try using other objects. Anything that has two ‘states’ is suitable. For example, you could use playing cards, coins (heads or tails) or cards with 0 or 1 printed on them (to relate to the binary system).
2. What happens if two, or more, cards are flipped? (It is not always possible to know exactly which two cards were flipped, although it is possible to tell that something has been changed. You can usually narrow it down to one of two pairs of cards. With 4 flips it is possible that all the parity bits will be correct afterwards, and so the error could go undetected.)
3. Try this with a much larger layout e.g. 9×9 cards, with the extra row and column expanding it to 10×10 . (It will work for any size layout, and doesn’t have to be square).
4. Another interesting exercise is to consider the lower right-hand card. If you choose it to be the correct one for the column above, then will it be correct for the row to its left? (The answer is yes, always, if you use even parity.)
5. In this card exercise we have used even parity—using an even number of coloured cards. Can we do it with odd parity? (This is possible, but the lower right-hand card only works out the same for its row and column if the numbers of rows and columns are both even or both odd. For example, a 5×9 layout will work fine, or a 4×6 , but a 3×4 layout won’t.)

A Real-Life Example for Experts!

This same checking technique is used with book codes and bar codes. Published books have a ten- or 13-digit code usually found on the back cover. The last digit is a check digit, just like the parity bits in the exercise.

This means that if you order a book using its ISBN (International Standard Book Number), the website can check that you haven't made a mistake. They simply look at the checksum. That way you don't end up waiting for the wrong book!

Here's how to work out the checksum for a 10-digit book code:

Multiply the first digit by ten, the second by nine, the third by eight, and so on, down to the ninth digit multiplied by two. Each of these values is then added together.

For example, the ISBN 0-13-911991-4 gives a value

$$\begin{aligned} & (0 \times 10) + (1 \times 9) + (3 \times 8) + (9 \times 7) + (1 \times 6) \\ & + (1 \times 5) + (9 \times 4) + (9 \times 3) + (1 \times 2) \\ & = 172 \end{aligned}$$

Then divide your answer by eleven. What is the remainder?

$$172 \div 11 = 15 \text{ remainder } 7$$

If the remainder is zero, then the checksum is zero, otherwise subtract the remainder from 11 to get the checksum.

$$11 - 7 = 4$$

Look back. Is this the last digit of the ISBN? Yes!

If the last digit of the ISBN wasn't a four, then we would know that a mistake had been made.

It is possible to come up with a checksum of the value of 10, which would require more than one digit. When this happens, the character X is used.

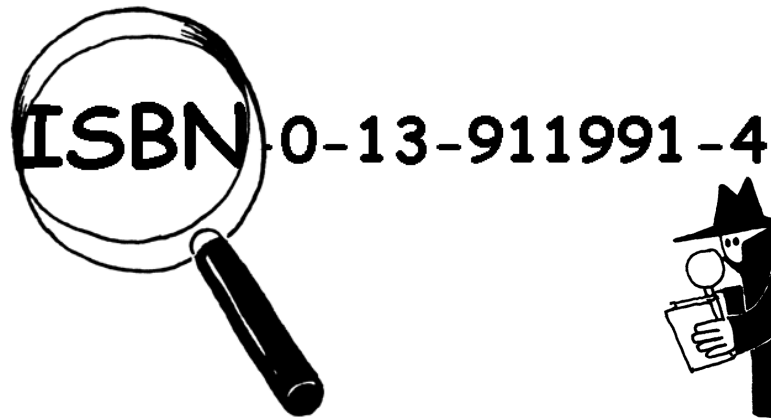


▲ A barcode (UPC) from a box of Weet-Bix™

Another example of the use of a check digit is the bar codes on grocery items. This uses a different formula (the same formula is used for 13-digit book codes). If a bar code is misread the final digit should be different from its calculated value. When this happens the scanner beeps and the checkout operator re-scans the code. Check digits are also used for bank account numbers, social security numbers, tax numbers, numbers on trains and rolling stock, and many other applications where people are copying a number and need some assurance that it has been typed in correctly.

Check that book!

**Detective Blockbuster
Book Tracking Service, Inc.**



We find and check ISBN checksums for a small fee.

Join our agency—look in your classroom or library for real ISBN codes.

Are their checksums correct?

Sometimes errors are made.

Some of the common errors are:

- ✗ a digit has its value changed;
- ✗ two adjacent digits are swapped with each other;
- ✗ a digit is inserted in the number; and
- ✗ a digit is removed from the number

Can you find a book with the letter X for a checksum of 10? It shouldn't be too hard to find—one in every 11 should have it.

What sort of errors might occur that wouldn't be detected? Can you change a digit and still get the correct checksum? What if two digits are swapped (a common typing error)?

What's it all about?

Imagine you are depositing \$10 cash into your bank account. The teller types in the amount of the deposit, and it is sent to a central computer. But suppose some interference occurs on the line while the amount is being sent, and the code for \$10 is changed to \$1,000. No problem if you are the customer, but clearly a problem for the bank!

It is important to detect errors in transmitted data. So a receiving computer needs to check that the data coming to it has not been corrupted by some sort of electrical interference on the line. Sometimes the original data can be sent again when an error has been transmitted, but there are some situations when this is not feasible, for example if a disk has been corrupted by exposure to magnetic or electrical radiation, by heat or by physical damage. If data is received from a deep space probe, it would be very tedious to wait for retransmission if an error had occurred! (It takes just over half an hour to get a radio signal from Jupiter when it is at its closest to Earth!)

We need to be able to recognize when the data has been corrupted (*error detection*) and to be able to reconstruct the original data (*error correction*).

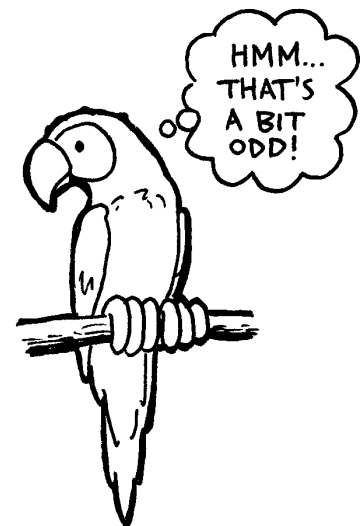
The same technique as was used in the “card flip” game is used on computers. By putting the bits into imaginary rows and columns, and adding parity bits to each row and column, we can not only detect if an error has occurred, but *where* it has occurred. The offending bit is changed back, and so we have performed error correction.

Of course computers often use more complex error control systems that are able to detect and correct multiple errors. The hard disk in a computer has a large amount of its space allocated to correcting errors so that it will work reliably even if parts of the disk fail. The systems used for this are closely related to the parity scheme.

And to finish, a joke that is better appreciated after doing this activity:

Q: What do you call this: “Pieces of nine, pieces of nine”?

A: A parrotty error.



Solutions and hints

Errors that would not be detected by an ISBN-10 checksum are those where one digit increases and another decreases to compensate. Then the sum might still be the same. However, because of the way the calculation is done, this is unlikely to happen. In other systems (such as ISBN-13) there are other types of errors that might not be detected, such as three consecutive digits being reversed, but most of the common errors (typing one digit incorrectly, or swapping two adjacent digits) will be detected.